

COMP527 - CA Assignment 2 - Evaluation

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Data

The clustering has the classes red, purple and blue. A total of 27 data points are distributed as follows:

Cluster	Red	Purple	Blue	Total
Cluster 1	4	3	3	10
Cluster 2	2	2	5	9
Cluster 3	2	4	2	8
Total	8	9	10	27

Table 1: Composition of each cluster by class.

Clusters

Each cluster is matched to its majority class: Cluster 1 to red (4 points), Cluster 2 to blue (5 points), and Cluster 3 to purple (4 points).

The confusion matrix is as follows:

	Red	Purple	Blue
Cluster 1	4	3	3
Cluster 2	2	2	5
Cluster 3	2	4	2

Table 2: Confusion matrix.

1 Macro-averaged Precision, Recall, and F-score

For each cluster, precision is the proportion of points in that cluster belonging to its majority class, and recall is the proportion of all points of that class captured by the cluster. The F-score is their harmonic mean.

Cluster 1 (majority class: red)

$$P_1 = \frac{4}{10} = 0.4000 \quad R_1 = \frac{4}{8} = 0.5000 \quad F_1 = \frac{2 \times 0.4 \times 0.5}{0.4 + 0.5} = 0.4444$$

Cluster 2 (majority class: blue)

$$P_2 = \frac{5}{9} = 0.5556 \quad R_2 = \frac{5}{10} = 0.5000 \quad F_2 = \frac{2 \times 0.5556 \times 0.5}{0.5556 + 0.5} = 0.5263$$

Cluster 3 (majority class: purple)

$$P_3 = \frac{4}{8} = 0.5000 \quad R_3 = \frac{4}{9} = 0.4444 \quad F_3 = \frac{2 \times 0.5 \times 0.4444}{0.5 + 0.4444} = 0.4706$$

Macro averages

$$\begin{aligned} \bar{P} &= \frac{0.4000 + 0.5556 + 0.5000}{3} = 0.4852 \\ \bar{R} &= \frac{0.5000 + 0.5000 + 0.4444}{3} = 0.4815 \\ \bar{F} &= \frac{0.4444 + 0.5263 + 0.4706}{3} = 0.4804 \end{aligned}$$

2 B-CUBED Precision, Recall, and F-score

B-CUBED computes precision and recall at the individual point level. For a point i in cluster C with class L :

$$P(i) = \frac{|\{j \in C : L(j) = L(i)\}|}{|C|} \quad R(i) = \frac{|\{j \in C : L(j) = L(i)\}|}{|\{j : L(j) = L(i)\}|}$$

The overall B-CUBED scores are the averages over all points. The per-class contributions are shown below:

Class	Cluster	n	Same in cluster	Total in class	P	R
Red	1	4	4	8	0.4000	0.5000
Red	2	2	2	8	0.2222	0.2500
Red	3	2	2	8	0.2500	0.2500
Purple	1	3	3	9	0.3000	0.3333
Purple	2	2	2	9	0.2222	0.2222
Purple	3	4	4	9	0.5000	0.4444
Blue	1	3	3	10	0.3000	0.3000
Blue	2	5	5	10	0.5556	0.5000
Blue	3	2	2	10	0.2500	0.2000

Table 3: Per-point B-CUBED precision and recall contributions by class and cluster.

Each row's P and R values are weighted by n (the number of points contributing that value). The overall averages across all 27 points are:

$$P_{B3} = \frac{4(0.4) + 2(0.2222) + 2(0.25) + 3(0.3) + 2(0.2222) + 4(0.5) + 3(0.3) + 5(0.5556) + 2(0.25)}{27} = 0.3728$$

$$R_{B3} = \frac{4(0.5) + 2(0.25) + 2(0.25) + 3(0.3333) + 2(0.2222) + 4(0.4444) + 3(0.3) + 5(0.5) + 2(0.2)}{27} = 0.3712$$

$$F_{B3} = \frac{2 \times 0.3728 \times 0.3712}{0.3728 + 0.3712} = 0.3720$$

Thus, for Macro-averaged values: Precision = 0.4852, Recall = 0.4815 and F-score: 0.4804.

And for B-CUBED, Precision = 0.3728, Recall = 0.3712 and F-score = 0.3720